

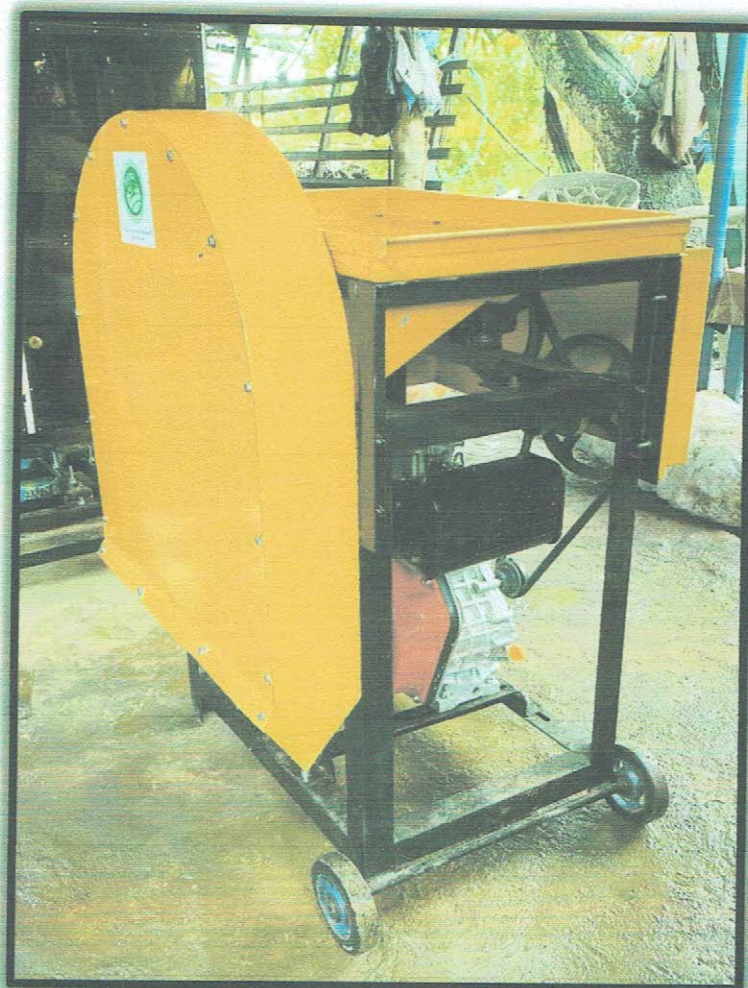


AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0012



Agricultural Machinery – Chipping Machine ACPF CHIM22

Test Report Number	2023 – 0012
Date of Issuance	January 28, 2023
Valid Until	January 28, 2028
Agricultural Machinery	Chipping Machine
Type	Flywheel, Power-operated
Brand	ACPF
Model	CHIM22
Test Requested by	ACPF AGRICULTURAL MACHINERY TRADING San Juan, Libmanan, Camarines Sur

SUMMARY

Performance Criteria	Requirement as per PAES 222:2005	AMTEC Test
Thickness setting	At least three	None
Coefficient of Variation of Chip Thickness, %, maximum	5	47.91
Quality of Output	Clear-cut chipped material	Cut chipped material
Noise Level, dB(A), maximum	92 ^a	99.9

^a Allowable noise level for six (6) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines. 1983.

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Not provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly, and installation	Not provided
h. Specifications	Not provided
i. Dismantling and disposal	Not provided
j. Warranty	Not provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	ACPF AGRICULTURAL MACHINERY TRADING
4. Prime mover used	8.95 kW YAMASAKI Diesel Engine
5. Number of operators	Two (One feeder, one bagger)
6. Loading of input	Input materials are placed in a feeding table and manually loaded into the input chute.
7. Cleaning of parts	All parts are accessible for cleaning. The chipping blade can be cleaned by removing the chipping assembly cover.
8. Adjusting and repairing of parts	The speed of the chipping blade can be adjusted by adjusting the throttle control of the prime mover. Parts can be repaired by removing the belt cover and/or the chipping assembly cover.
9. Collecting of output	A plastic mat is placed below the output chute to collect the output materials.
10. Transporting the machine	The machine is equipped with transport wheels and push handles for transport.
11. Safety features	The transmission system and chipping assembly were provided with covers made of G.I sheet.
12. Failures or abnormalities that may be observed on the machine or its component parts during and after the chipping operation	There were no failures or abnormalities observed on the machine during and after the chipping operation.

OBSERVATIONS (Cont'n)

13. Steel bars and metal sheet or plate shall be generally used in the manufacture of the different components of chipping machine. However, for parts which are in direct contact with the materials to be chipped, food grade materials shall be used.	The machine is generally made of steel bars and metal sheets. The chipping blade is made up of cold rolled steel.
14. Chipping elements should be made of either AISI 1040 to AISI 1045 or its ISO equivalent.	Not verified.
15. To safeguard against contact with cutting mechanism, the machine shall have a safety distance depending on its feed opening size and feed opening shape as specified in Tables 1 and 2 of PAES 222:2005.	The minimum distance of the feeder's hand to the cutting mechanism is 160 mm. The minimum requirement for a slot more than 50 mm and less than 250 mm is 850 mm.
16. The discharge chute shall be designed to prevent direct access to and accidental contact with the cutting elements.	The discharge chute is designed to prevent direct access to and accidental contact with the cutting elements.
17. The chipping machine shall be free from manufacturing defects that may be detrimental to its operation.	There were no manufacturing defects observed that may be detrimental to its operation.
18. The chipping machine shall be free from sharp edges and surfaces (except for the cutting edge) that may injure the operator.	There were no sharp edges and surfaces observed.
19. Guards allowing access to the cutting chamber shall be interlocked to cause the moving parts to come to rest before access can be gained. Other guards (i.e. belt cover) shall be permanently attached to the machine and shall not be detachable without the use of tools.	The cover of the chipping assembly is not interlocked with the cutting chamber. The transmission assembly cover can only be removed using tools.
20. Rotating parts should be dynamically balanced.	Not verified
21. Mechanism for immediate disengagement of power transmission shall be provided.	There was no mechanism for immediate disengagement of power transmission provided.

OBSERVATIONS (Cont'n)

22. Sealed type bearings should be used as protection against dust. There shall be provision for lubrication of non-sealed type bearings and bushings.	Sealed type bearings were used on the machine.
23. Belt tightening and adjustments shall be provided.	There was no provision for belt tightening and adjustments provided.
24. Any uncoated metallic surfaces shall be free from rust and shall be painted properly.	All uncoated metallic surfaces were free from rust.
25. Other remarks	The machine did not produce distinctively clear-cut chipped materials.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	January 19, 2023
Location of Test	Binobong, Pili, Camarines Sur 13°31'54.30" N, 123°15'37.73" E
Items Tested	Operator's Manual, Chip Thickness, Speed of Components, Chipping Capacity, Chipping Efficiency, Fuel Consumption Rate, Noise Level, and Number of Operators

METHODS OF TEST

The chipping machine was tested based on the PAES 223:2004 Agricultural Machinery – Chipping Machine – Methods of Test.

DESCRIPTION OF THE MACHINE

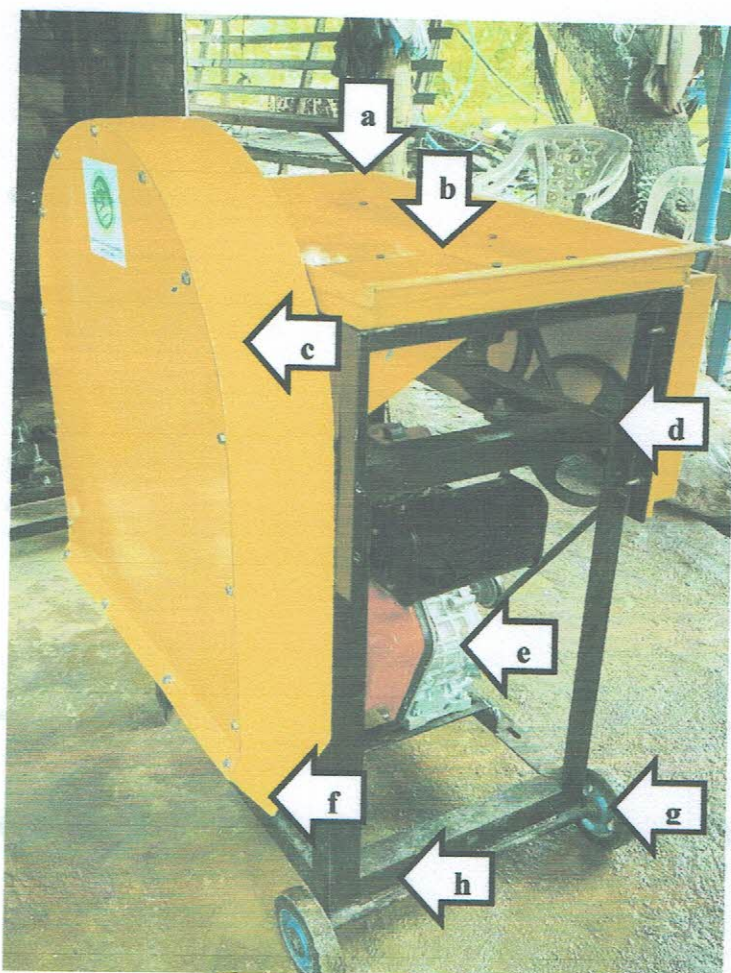


Figure 1. Parts of the ACPF CHIM22 Chipping Machine powered by 8.95 kW YAMASAKI Diesel Engine include the following: (a) Feeding table, (b) Feed opening slot, (c) Chipping chamber, (d) Belt-pulley transmission system, (e) Prime mover, (f) Outlet chute (g) Transport wheel, and (h) Mainframe.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	1000	998
1.1.2	Width	700	726
1.1.3	Height	1200	1209
1.2	Weight without prime mover, kg	ND	NM
2	Feeding slot		
2.1	Dimension, L × W, mm	320 × 210	317 × 206
2.2	Height from the ground, mm	1050	1076
2.3	Material of construction	G.I Sheet	G.I Sheet (1.0 mm)
3	Chipping assembly		
3.1	Chipping plate		
3.1.1	Diameter, mm	635	608
3.1.2	Thickness, mm	3	4.4
3.1.3	Material of construction	MS Plate	MS Plate
3.2	Chipping blade		
3.2.1	Type	ND	Straight edge
3.2.2	Dimension, L × W × t, mm	200 × 60 × 5.0	205 × 62 × 8.0
3.2.3	Width of blade opening, mm	ND	19
3.2.4	Shape	ND	Rectangular
3.2.5	Number of blades	3	3
3.2.6	Means of attachment	ND	Bolted
3.2.7	Material of construction	Cold rolled steel	ND
4	Output chute		
4.1	Material of construction	G.I Sheet	G.I Sheet
4.2	Dimensions, L × W, mm	690 × 130	NM
4.3	Dimensions of opening, L × W × t, mm	690 × 115 × 1.2	681 × 105 × NM
4.4	Height from the ground, mm	310	333

^a The data were obtained from the specification sheet provided by the requesting party.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item	Manufacturer's Specification ^a	AMTEC Verification
5 Engine		
5.1 Brand	YAMASAKI	YAMASAKI
5.2 Model	BYK188FA	BYK188FA
5.3 Type	Diesel engine	Single vertical cylinder, air-cooled, diesel engine
5.4 Make or manufacturer	China	China
5.5 Serial number	BTY20150420016	BTY20150420016
5.6 Rated power, kW	8.95	NM
5.7 Rated speed, rpm	3600	NM
5.8 Displacement, cm ³	418	NM
5.9 Fuel type	ND	Diesel
5.10 Fuel tank capacity, L	ND	NM
5.11 Cooling system	Air-cooled	Air cooled
5.12 Starting system	ND	Rope-recoil starter
5.13 Weight, kg	45	NM
6 Power transmission system		
6.1 Prime mover to Intermediate shaft	ND	V-belt and pulley
6.1.1 Prime mover ^b	ND	76.2 × 1B × 25
6.1.2 Intermediate shaft ^b	ND	254.0 × 1B × 25
6.1.3 Belt size	B-36	B-36
6.2 Intermediate shaft to chipping assembly	ND	V-belt and pulley
6.2.1 Intermediate shaft ^b	ND	76.2 × 1B × 25
6.2.2 Chipping assembly ^b	ND	228.0 × 1B × 25
6.2.3 Belt size	B-35	B-29
7 Trailing wheel, D × W, mm	ND	150 × 35
8 Main frame, L × W, mm	ND	685 × 482
9 Safety devices	ND	Belt cover
10 Special features	ND	None

^a The data were obtained from specification sheet provided by the requesting party.

^b Pulley diameter, mm × number of belts × shaft diameter, mm

Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Conditions of crop		
1.1	Crop	Unpeeled cassava tubers	
1.2	Variety	ND	
1.3	Source	Tinangis, Pili, Camarines Sur	
1.4	Moisture content, % _{wb}	59.80	
1.5	Dimensions, L × D, mm	205 × 56.40	
2	Performance test		
2.1	Weight of input, kg	100.1	
2.2	Chipping capacity, kg/h	1792.08	
2.3	Chipping efficiency, %	93.11	
2.4	Speed of components, rpm	Without load	With load
2.4.1	Engine shaft	2801	2777
2.4.2	Intermediate shaft	863	852
2.4.3	Chipping shaft	297	289
2.5	Noise level, dB(A)	Without load	With load
2.5.1	Feeding operator's ear level	97.8	99.9
2.5.2	Bagger's ear level	97.4	98.5
2.6	Fuel consumption rate, L/h	0.366	
3	Laboratory analysis		
3.1	Average thickness, mm	9.58	
3.2	Coefficient of variation, %	47.91	
3.3	Percent error ^a , %	54.24	

Note: ND – No Data

^a – Based on the thickness setting of 4.5 mm as target for percent error computation.

OBSERVATIONS

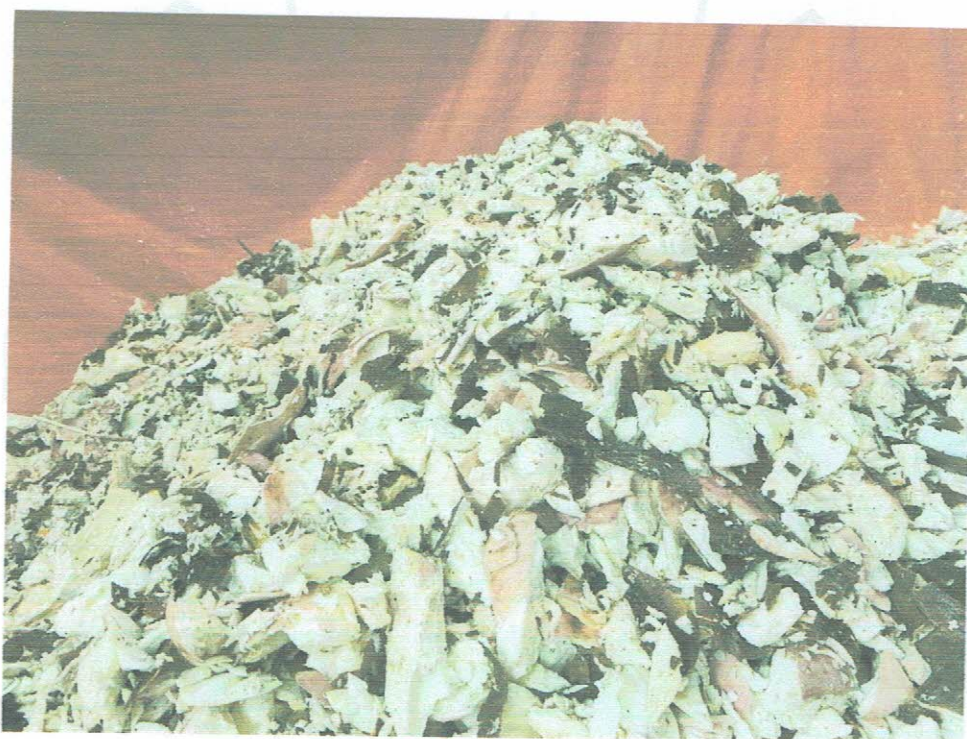
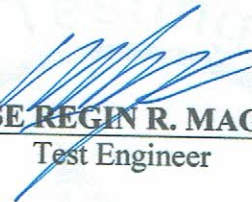
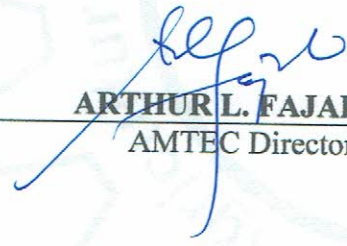


Figure 2. Quality of output

Tested by:
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AMTEC Director**FEB 01 2023**
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